Assignment 8: Basic SQL Queries and Aggregate Functions

COLLEGE MANAGEMENT SYSTEM

DBMS Assignment 8

Database: CollegeManagementSystem

SQL Platform: MySQL 8.0.16 or later

Project scope: departments, faculty, students, courses, and enrollments.

Note: Query outputs described in this report are expected results from the included fictional sample data. Run the SQL script in MySQL Workbench and capture actual screenshots if your instructor requires them.

WHERE and ORDER BY

```
SELECT student_name,register_no FROM Students
WHERE department_id=1 ORDER BY student_name;
```

GROUP BY

```
SELECT department_id,COUNT(*) AS student_count
FROM Students GROUP BY department_id;
```

HAVING

```
SELECT course_id,COUNT(*) AS enrolled_students
FROM Enrollments GROUP BY course_id HAVING COUNT(*)>=2;
```

Aggregate functions

```
SELECT COUNT(*) AS total_students FROM Students;
SELECT SUM(credits) AS total_credits FROM Courses;
SELECT AVG(salary) AS average_salary FROM Faculty;
SELECT MIN(salary) AS min_salary,MAX(salary) AS max_salary FROM Faculty;
```

Expected outputs and explanations

Computer Science student list sorted by name: Aarav Sharma, Diya Nair, Ishaan Patel. Student counts by department_id: 1 = 3, 2 = 2, 3 = 1. Course IDs with at least two enrollments: course 1 = 3, course 2 = 2, course 4 = 2. Total students = 6. Sum of course credits = 18. Average faculty salary = 61,250. Minimum salary = 58,000; maximum salary = 65,000. WHERE filters rows; ORDER BY sorts rows; GROUP BY creates groups; HAVING filters groups. COUNT, SUM, AVG, MIN, and MAX calculate count, total, mean, minimum, and maximum.

End of Assignment Report